Solutions

Maths

Q1. If x_0 is a real solution of the equation

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1}(x+1)$$

then the value of $2x_0 + 1$ is:

(A) $\sqrt{5}$ (B) $-\sqrt{5}$ (C) $\sqrt{3}$ (D) $-\sqrt{3}$

Ans. B

Sol. We know the definition of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ splits into different intervals:

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} -\pi - 2\tan^{-1}x, & x < -1 \\ 2\tan^{-1}x, & -1 \le x \le 1 \\ \pi - 2\tan^{-1}x, & x > 1 \end{cases}$$

We analyze the given equation across these intervals: Case 1: $-1 \le x \le 1$ The equation becomes:

$$2\tan^{-1}x = 2\tan^{-1}(x+1) \implies x = x+1 \implies 0 = 1 \text{ (No solution)}$$

Case 2: $x > 1$ The equation becomes:

$$\pi - 2\tan^{-1}x = 2\tan^{-1}(x+1) \implies \tan^{-1}x + \tan^{-1}(x+1) = \frac{\pi}{2}$$

Since $x > 1$, both arguments are positive, so:

$$x(x+1) = 1 \implies x^2 + x - 1 = 0 \implies x = \frac{-1 \pm \sqrt{5}}{2}$$

However, $\frac{-1+\sqrt{5}}{2} \approx 0.618 < 1$ and the negative root is less than zero. Both fail the condition $x > 1$. Thus, no solution in this interval. Case 3: $x < -1$ The equation becomes:

$$-\pi - 2\tan^{-1}x = 2\tan^{-1}(x+1) \implies \tan^{-1}x + \tan^{-1}(x+1) = -\frac{\pi}{2}$$

Since $x < -1$, both x and $x+1$ are negative. Let $x = -y$ where $y > 1$:

$$\tan^{-1}(-y) + \tan^{-1}(-(y-1)) = -\frac{\pi}{2} \implies \tan^{-1}y + \tan^{-1}(y-1) = \frac{\pi}{2}$$

Since $y > 1$, both terms are positive, giving:

$$y(y-1) = 1 \implies y^2 - y - 1 = 0 \implies y = \frac{1+\sqrt{5}}{2} \text{ (as } y > 1)$$

Substituting back for x:

$$x_0 = -y = \frac{-1-\sqrt{5}}{2}$$

This satisfies $x < -1$ (≈ -1.618). Thus, the unique real solution is $x_0 = \frac{-1-\sqrt{5}}{2}$. The value of $2x_0 + 1$ is:

$$2\left(\frac{-1-\sqrt{5}}{2}\right) + 1 = -1 - \sqrt{5} + 1 = -\sqrt{5}$$

Correct Option: (B)

Q2. Let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x and $\{x\}$ denote the fractional part of x. Then the value of the limit

$$\lim_{x \to 0} \frac{\sin^2 x}{\{x\}^2 + \lfloor x \rfloor (2x+1)}$$

is:

(A) 0 (B) 1 (C) -1 (D) does not exist

Ans. B

Sol. To evaluate the limit, we compute the Right-Hand Limit (RHL) and Left-Hand Limit (LHL) as $x \to 0$.

Right-Hand Limit (RHL):

For $x \to 0^+$, $x \in (0, 1)$, which implies:

$$\lfloor x \rfloor = 0 \quad \text{and} \quad \{x\} = x$$

Substituting these into the denominator gives:

$$\{x\}^2 + \lfloor x \rfloor (2x+1) = x^2 + 0 \cdot (2x+1) = x^2$$

Thus, the RHL is:

$$\lim_{x \to 0^+} \frac{\sin^2 x}{x^2} = 1$$

Left-Hand Limit (LHL):

For $x \to 0^-$, $x \in (-1, 0)$, which implies:

$$\lfloor x \rfloor = -1 \quad \text{and} \quad \{x\} = x - \lfloor x \rfloor = x + 1$$

Substituting these into the denominator gives:

$$\{x\}^2 + \lfloor x \rfloor (2x+1) = (x+1)^2 + (-1)(2x+1) = x^2 + 2x + 1 - 2x - 1 = x^2$$

Thus, the LHL is:

$$\lim_{x \to 0^-} \frac{\sin^2 x}{x^2} = 1$$

Since RHL $=$ LHL $= 1$, the limit exists and is equal to 1.
Correct Option: (B)

Q3. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$f(x) = \begin{cases} ae^x + b|x-1| & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ \frac{\sin x + c\ln(1+x)}{x} & \text{if } x > 0 \end{cases}$$

where a, b, c are real constants. If $f(x)$ is differentiable at $x = 0$, then the value of $4a + 4b + c$ is:

(A) 7 (B) 9 (C) 11 (D) 13

Ans. B

Sol. For $f(x)$ to be differentiable at $x = 0$, it must first be continuous at $x = 0$.Thus, we must have:

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = f(0)$$

For $x < 0$, we have $|x - 1| = 1 - x$. Therefore :

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (ae^x + b(1-x)) = a + b$$

For $x > 0$:

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{\sin x + c\ln(1+x)}{x} = 1 + c$$

Equating these to $f(0) = 2$:

$$a + b = 2 \quad \text{--- } (1)$$

$$1 + c = 2 \implies c = 1 \quad \text{--- } (2)$$

\nNow, checking the differentiability at $x = 0$:\nThe left-hand derivative (LHD) is given by:

$$\text{LHD} = \lim_{x \to 0^-} \frac{d}{dx}(ae^x + b - bx) = \lim_{x \to 0^-} (ae^x - b) = a - b$$

The right-hand derivative (RHD) using expansion for $c = 1$:

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$\ln(1+x) = x - \frac{x^2}{2} + o(x^2)$$

$$\implies f(x) = \frac{\left(x - \frac{x^3}{6}\right) + \left(x - \frac{x^2}{2}\right)}{x} = 2 - \frac{x}{2} - \frac{x^2}{6}$$

Thus, the right-hand derivative at $x = 0$ is:

$$\text{RHD} = \lim_{x \to 0^+} \frac{f(x) - f(0)}{x} = \lim_{x \to 0^+} \frac{\left(2 - \frac{x}{2} - \frac{x^2}{6}\right) - 2}{x} = -\frac{1}{2}$$

Equating LHD and RHD for differentiability:

$$a - b = -\frac{1}{2} \quad \text{--- (3)}$$

Solving equations (1) and (3) simultaneously:

$$2a = \frac{3}{2} \implies a = \frac{3}{4}$$

$$2b = \frac{5}{2} \implies b = \frac{5}{4}$$

We need to find the value of $4a + 4b + c$:

$$4a + 4b + c = 4\left(\frac{3}{4}\right) + 4\left(\frac{5}{4}\right) + 1 = 3 + 5 + 1 = 9$$

Hence, the correct option is (B).

Q4. Let $S = \{1, 2, 3, 4, 5\}$. Two subsets A and B of S are chosen at random and independently from the power set of S. Given that the intersection $A \cap B$ contains an odd number of elements, the probability that $A \cup B = S$ is

(A) $\frac{121}{496}$ (B) $\frac{121}{512}$ (C) $\frac{61}{248}$ (D) $\frac{125}{496}$

Ans. A

Sol. The total number of ways to choose two subsets A and B from a set of 5 elements is $2^5 \times 2^5 = 1024$. Each pair of subsets (A, B) is equally likely. Let F be the event that the intersection $A \cap B$ contains an odd number of elements, and let E be the event that $A \cup B = S$. We need to find the conditional probability $P(E|F) = \frac{n(E \cap F)}{n(F)}$. Let $|A \cap B| = k$, where $k \in \{1, 3, 5\}$. To form such pairs of subsets, we first choose k elements out of 5 to be in $A \cap B$ in $\binom{5}{k}$ ways. For each of the remaining $5 - k$ elements, it can belong to exactly one of three regions: only in A, only in B, or in neither A nor B. This gives 3^{5-k} choices. Thus, the total

number of favorable ways for event F is:

$$n(F) = \sum_{k\in\{1,3,5\}} \binom{5}{k} 3^{5-k}$$

$$n(F) = \binom{5}{1}3^4 + \binom{5}{3}3^2 + \binom{5}{5}3^0 = 5(81) + 10(9) + 1(1) = 405 + 90 + 1 = 496$$

Alternatively, using binomial expansions:

$$\sum_{k\text{ odd}} \binom{5}{k} 3^{5-k} = \frac{(3+1)^5 - (3-1)^5}{2} = \frac{1024 - 32}{2} = 496$$

For the event $E \cap F$ to occur, we must have $A \cup B = S$ along with $|A \cap B| = k$ being odd. The condition $A \cup B = S$ implies that no element can belong to the region "neither A nor B". Therefore, for each of the remaining $5 - k$ elements, there are only 2 choices: either only in A or only in B. Thus, the number of outcomes in $E \cap F$ is:

$$n(E \cap F) = \sum_{k\in\{1,3,5\}} \binom{5}{k} 2^{5-k}$$

$$n(E \cap F) = \binom{5}{1}2^4 + \binom{5}{3}2^2 + \binom{5}{5}2^0 = 5(16) + 10(4) + 1(1) = 80 + 40 + 1 = 121$$

Alternatively, using binomial expansions:

$$\sum_{k\text{ odd}} \binom{5}{k} 2^{5-k} = \frac{(2+1)^5 - (2-1)^5}{2} = \frac{243 - 1}{2} = 121$$

$$P(E|F) = \frac{n(E \cap F)}{n(F)} = \frac{121}{496}$$

Hence, the correct option is (A).

Q1. Let $\mathbb{R}$ denote the set of all real numbers and $\mathbb{Z}$ denote the set of all integers. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by: $f(x) = \begin{cases} (1-x^2)\left(2 + \sin\left(\frac{\pi}{1-x}\right)\right) & ,x < 1 \& x \notin Z \\ e^{-\frac{1}{(x-1)^2}} \sin\left(\frac{1}{\{x\}}\right) & ,x > 1 \text{and} x \notin Z \text{ where } \{x\} \\ 0 & ,x \in Z \end{cases}$ denotes the fractional part function of x. Which of the following statements is (are) **TRUE**?

(A) f is continuous at $x = 1$ but not differentiable at $x = 1$

(B) f is differentiable at $x = -1$ and $f'(-1) = 6$ (C) $\lim\limits_{x\to 1^+} f'(x) = 0$

(D) f is continuous at $x = 0$

Ans. A, B, C

Sol. Analysis for Option (A): To check continuity at $x = 1$, we evaluate the left-hand limit (LHL) and right-hand limit (RHL):

$$\text{LHL} = \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (1-x^2)\left(2 + \sin\left(\frac{\pi}{1-x}\right)\right) = 0$$

Since $\left|2+\sin\left(\frac{\pi}{1-x}\right)\right| \leq 3$ is bounded and $(1-x^2) \to 0$.\nFor $1<x<2$, we have $\{x\}=x-1$, so:

$$\text{RHL} = \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} e^{-\frac{1}{(x-1)^2}} \sin\left(\frac{1}{x-1}\right) = 0$$

Since $e^{-\frac{1}{(x-1)^2}} \to 0$ and $\sin\left(\frac{1}{x-1}\right)$ is bounded.\nSince $\text{LHL} = \text{RHL} = f(1) = 0$, f is continuous at $x=1$.\nFor differentiability at $x=1$:

$$\text{LHD} = \lim_{x\to 1^-} \frac{f(x)-f(1)}{x-1} = \lim_{x\to 1^-} \frac{(1-x)(1+x)\left(2+\sin\left(\frac{\pi}{1-x}\right)\right)}{x-1} = \lim_{x\to 1^-} -(1+x)\left(2+\sin\left(\frac{\pi}{1-x}\right)\right.$$

As $x \to 1^-$, $\sin\left(\frac{\pi}{1-x}\right)$ oscillates infinitely between -1 and 1, so the limit does not exist. Thus, f is not differentiable at $x=1$. Hence, option (A) is correct.\n\nAnalysis for Option (B):At $x=-1$, which is an integer, $f(-1)=0$. In a deleted neighborhood of $x=-1$, $x<1$ and x is not an integer, so $f(x)=(1-x^2)\left(2+\sin\left(\frac{\pi}{1-x}\right)\right)$. Evaluating the derivative using the definition:

$$f'(-1) = \lim_{x\to -1} \frac{f(x)-f(-1)}{x-(-1)} = \lim_{x\to -1} \frac{(1-x)(1+x)\left(2+\sin\left(\frac{\pi}{1-x}\right)\right)}{1+x} = \lim_{x\to -1} (1-x)\left(2+\sin\left(\frac{\pi}{1-x}\right.\right.$$

\n

$$f'(-1) = (1-(-1))\left(2+\sin\left(\frac{\pi}{1-(-1)}\right)\right) = 2\left(2+\sin\left(\frac{\pi}{2}\right)\right) = 2(2+1) = 6$$

Thus, f is differentiable at $x=-1$ and $f'(-1)=6$. Option (B) is correct.\n\nAnalysis for Option (C): For $x \in (1,2)$, $f(x) = e^{-\frac{1}{(x-1)^2}} \sin\left(\frac{1}{x-1}\right)$. Let $u = x-1 \to 0^+$.

$$f(u) = e^{-u^{-2}} \sin(u^{-1})$$

\n

$$f'(u) = 2u^{-3}e^{-u^{-2}}\sin(u^{-1}) - u^{-2}e^{-u^{-2}}\cos(u^{-1})$$

Substituting $t = u^{-1}t \to \infty$ as $u \to 0^+$:

$$\lim_{x\to 1^+} f'(x) = \lim_{t\to\infty} \left(2t^3 e^{-t^2}\sin(t) - t^2 e^{-t^2}\cos(t)\right)$$

Since $\lim_{t\to\infty} \frac{t^3}{e^{-t^2}} = 0$ and $\lim_{t\to\infty} \frac{t^2}{e^{-t^2}} = 0$ while $\sin(t)$ and $\cos(t)$ are bounded, the limit evaluates to 0. Option (C) is correct.\n\nAnalysis for Option (D): Since $0 \in \mathbb{Z}$, $f(0)=0$. However, checking the limit as $x \to 0$:

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} (1-x^2)\left(2+\sin\left(\frac{\pi}{1-x}\right)\right) = (1)\left(2+\sin(\pi)\right) = 2$$

Since $\lim_{x\to 0} f(x) = 2 \neq f(0)$, f is discontinuous at $x=0$. Option (D) is incorrect.

Q2. Let $[x]$ denote the greatest integer less than or equal to x. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$f(x) = [x]|x^2 - 4x + 3| \sin(\pi x)$$

Then which of the following statements is/are **TRUE**?

(A) f is continuous on $\mathbb{R}$. (B) f is differentiable at $x = 1$ and $x = 3$.

(C) The left-hand derivative of f at $x = 2$ is π and the right-hand derivative of f at $x = 2$ is 2π.

(D) The set of points where f is not differentiable is $\mathbb{Z} - \{1, 3\}$, where $\mathbb{Z}$ is the set of all integers.

Ans. A, B, C, D

Sol. Analysis of Continuity:

For any non-integer x, $[x]$ is locally constant, and $|x^2 - 4x + 3|$ and $\sin(\pi x)$ are continuous functions, so $f(x)$ is continuous.

For any integer $k \in \mathbb{Z}$:

$$\lim_{x \to k^-} f(x) = (k-1)|k^2 - 4k + 3| \cdot 0 = 0$$

$$\lim_{x \to k^+} f(x) = k|k^2 - 4k + 3| \cdot 0 = 0$$

Since $f(k) = 0$, we have $\lim_{x \to k} f(x) = f(k) = 0$. Hence, $f(x)$ is continuous for all $x \in \mathbb{R}$. Statement (A) is correct.

Analysis of Differentiability at Integers:

Let $k \in \mathbb{Z}$. The left-hand derivative (LHD) at $x = k$ is:

$$f'_-(k) = \lim_{h \to 0^+} \frac{f(k-h) - f(k)}{-h} = \lim_{h \to 0^+} \frac{(k-1)|(k-h-1)(k-h-3)| \sin(\pi(k-h))}{-h}$$

Using $\sin(k\pi - \pi h) = (-1)^{k+1} \sin(\pi h) \approx (-1)^{k+1} \pi h$:

$$f'_-(k) = \lim_{h \to 0^+} \frac{(k-1)|(k-1)(k-3)|(-1)^{k+1}\pi h}{-h} = (-1)^k \pi (k-1)|(k-1)(k-3)|$$

The right-hand derivative (RHD) at $x = k$ is:

$$f'_+(k) = \lim_{h \to 0^+} \frac{f(k+h) - f(k)}{h} = \lim_{h \to 0^+} \frac{k|(k+h-1)(k+h-3)| \sin(\pi(k+h))}{h}$$

Using $\sin(k\pi + h\pi) = (-1)^k \sin(\pi h) \approx (-1)^k \pi h$:

$$f'_+(k) = \lim_{h \to 0^+} \frac{k|(k-1)(k-3)|(-1)^k \pi h}{h} = (-1)^k \pi k|(k-1)(k-3)|$$

For $f(x)$ to be differentiable at $x = k$, we must have $f'_-(k) = f'_+(k)$:

$$(-1)^k\pi(k-1)|(k-1)(k-3)| = (-1)^k\pi k|(k-1)(k-3)|$$

Case 1: $|(k-1)(k-3)| = 0 \implies k = 1$ or $k = 3$.
In this case, both LHD and RHD become 0, so $f(x)$ is differentiable at $x = 1$ and $x = 3$ with $f'(1) = f'(3) = 0$. Statement (B) is correct.

Case 2: $|(k-1)(k-3)| \neq 0 \implies k \in \mathbb{Z} - \{1, 3\}$.
Dividing both sides of the condition by $(-1)^k\pi|(k-1)(k-3)|$ yields $k - 1 = k \implies -1 = 0$, which is impossible.
Thus, $f(x)$ is not differentiable at any integer $k \in \mathbb{Z}$–$\{1, 3\}$. Since $|x^2 - 4x + 3|$ is differentiable at all non-integers, $f(x)$ is differentiable at all non-integers. Thus, the exact set of non-differentiable points is $\mathbb{Z}$–$\{1, 3\}$. Statement (D) is correct.

Evaluating Derivatives at $x = 2$:
Substitute $k = 2$ into the LHD and RHD expressions:

$$f'_-(2) = (-1)^2\pi(2-1)|(2-1)(2-3)| = \pi(1)(1) = \pi$$

$$f'_+(2) = (-1)^2\pi(2)|(2-1)(2-3)| = \pi(2)(1) = 2\pi$$

Since $f'_-(2) = \pi$ and $f'_+(2) = 2\pi$, Statement (C) is correct.

Correct Options: (A), (B), (C), (D)

Q3. Consider three bags B_1, B_2 and B_3. Bag B_1 contains 2 white and 3 black balls, bag B_2 contains 3 white and 2 black balls, and bag B_3 contains 4 white and 1 black ball. A fair six-sided die is rolled. Let D denote the number appearing on the top face of the die. If $D = 1$, bag B_1 is selected; if $D \in \{2, 3\}$, bag B_2 is selected; and if $D \in \{4, 5, 6\}$, bag B_3 is selected. A ball is then drawn at random from the selected bag. Which of the following statements is (are) **TRUE**?

(A) The probability that the drawn ball is white is $\frac{2}{3}$

(B) Given that the drawn ball is white, the probability that the die showed a number greater than 3 is $\frac{3}{5}$

(C) Given that the drawn ball is white, the probability that the die showed an even number is $\frac{11}{20}$

(D) Given that the drawn ball is black, the probability that the die showed an even number is $\frac{1}{2}$

Ans. A, B, C

Sol. The probabilities of selecting the bags are:

$$P(B_1) = P(D = 1) = \frac{1}{6}$$

$$P(B_2) = P(D \in \{2,3\}) = \frac{2}{6} = \frac{1}{3}$$

$$P(B_3) = P(D \in \{4,5,6\}) = \frac{3}{6} = \frac{1}{2}$$

Let W be the event that the drawn ball is white. The conditional probabilities are:

$$P(W|B_1) = \frac{2}{5}, \quad P(W|B_2) = \frac{3}{5}, \quad P(W|B_3) = \frac{4}{5}$$

Using the law of total probability:

$$P(W) = P(B_1)P(W|B_1) + P(B_2)P(W|B_2) + P(B_3)P(W|B_3)$$

$$P(W) = \frac{1}{6} \cdot \frac{2}{5} + \frac{1}{3} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{4}{5} = \frac{2}{30} + \frac{6}{30} + \frac{12}{30} = \frac{20}{30} = \frac{2}{3}$$

Hence, statement (A) is correct.

For statement (B), a die roll greater than 3 corresponds exactly to choosing bag B_3. By Bayes' Theorem:

$$P(D > 3|W) = P(B_3|W) = \frac{P(B_3 \cap W)}{P(W)} = \frac{\frac{1}{2} \cdot \frac{4}{5}}{\frac{2}{3}} = \frac{\frac{2}{5}}{\frac{2}{3}} = \frac{3}{5}$$

Hence, statement (B) is correct.

Let F be the event that the die shows an even number, i.e., $D \in \{2,4,6\}$.

$$P(F \cap W) = P(D = 2 \cap W) + P(D = 4 \cap W) + P(D = 6 \cap W)$$

$$P(F \cap W) = \frac{1}{6} \cdot \frac{3}{5} + \frac{1}{6} \cdot \frac{4}{5} + \frac{1}{6} \cdot \frac{4}{5} = \frac{3+4+4}{30} = \frac{11}{30}$$

The conditional probability is:

$$P(F|W) = \frac{P(F \cap W)}{P(W)} = \frac{\frac{11}{30}}{\frac{2}{3}} = \frac{11}{20}$$

Hence, statement (C) is correct.

Let W^c be the event that the drawn ball is black. Then $P(W^c) = 1 - P(W) = \frac{1}{3}$.

$$P(F \cap W^c) = P(D = 2 \cap W^c) + P(D = 4 \cap W^c) + P(D = 6 \cap W^c)$$

$$P(F \cap W^c) = \frac{1}{6} \cdot \frac{2}{5} + \frac{1}{6} \cdot \frac{1}{5} + \frac{1}{6} \cdot \frac{1}{5} = \frac{2+1+1}{30} = \frac{4}{30}$$

The conditional probability is:

$$P(F|W^c) = \frac{P(F \cap W^c)}{P(W^c)} = \frac{\frac{4}{30}}{\frac{1}{3}} = \frac{4}{10} = \frac{2}{5}$$

Since $\frac{2}{5} \neq \frac{1}{2}$, statement (D) is incorrect.

Q4. Consider a data set consisting of 7 positive integers $x_1, x_2, x_3, x_4, x_5, x_6, x_7$ such that $x_1 \leq x_2 \leq x_3 \leq x_4 \leq x_5 \leq x_6 \leq x_7$. The arithmetic mean, the median, and the unique mode of this data set are all integers, with the arithmetic mean being 6, the median being 6, and the unique mode being 8. Which of the following statements is/are **TRUE**?

(A) The total number of such distinct data sets is 5.

(B) The maximum possible range of such a data set is 7.

(C) The number of observations strictly less than the arithmetic mean can be 3.

(D) The least possible value of x_3 is 4.

Ans. C

Sol. Since the data set has 7 elements sorted in non-decreasing order, the median is the 4th element, so $x_4 = 6$.

The unique mode is 8, which is greater than the median, so 8 must appear at least twice among x_5, x_6, x_7.

If the frequency of 8 is exactly 2, then $x_6 = x_7 = 8$. For 8 to be the unique mode, all other elements must appear at most once, meaning they must be strictly distinct:

$x_1 < x_2 < x_3 < 6 < x_5 < 8$

Since x_5 is an integer, $x_5 = 7$. The total sum of the observations is $7 \times 6 = 42$. Thus,

$x_1 + x_2 + x_3 + 6 + 7 + 8 + 8 = 42 \quad \Rightarrow x_1 + x_2 + x_3 = 13$

Since x_1, x_2, x_3 are distinct positive integers less than 6, their maximum possible values are 3, 4, and 5, giving a maximum sum of 3 + 4 + 5 = 12, which contradicts the requirement of 13. Hence, the frequency of 8 cannot be 2.

If the frequency of 8 is 4 or more, the median x_4 would become 8, which is a contradiction. Thus, the frequency of 8 must be exactly 3, which implies $x_5 = x_6 = x_7 = 8$.

The sum equation now becomes :

$x_1 + x_2 + x_3 + 6 + 8 + 8 + 8 = 42 \quad \Rightarrow x_1 + x_2 + x_3 = 12$

where $1 \leq x_1 \leq x_2 \leq x_3 \leq 6$. We perform a case analysis on x_3:

- Case 1: $x_3 = 6 \Rightarrow x_1 + x_2 = 6$. The valid pairs (x_1, x_2) are (1, 5), (2, 4), and (3,3). This gives 3 distinct data sets:

1. {1,5,6,6,8,8,8}
2. {2,4,6,6,8,8,8}
3. {3,3,6,6,8,8,8}

- Case 2: $x_3 = 5 \Rightarrow x_1 + x_2 = 7$. Since $x_2 \le x_3 = 5$, the valid pairs are (2, 5) and (3, 4). This gives 2 distinct data sets:

4. {2,5,5,6,8,8,8}
5. {3,4,5,6,8,8,8}

- Case 3: $x_3 = 4 \Rightarrow x_1 + x_2 = 8$. With $x_2 \le 4$, the only pair is (4, 4), yielding the triplet (4, 4, 4). The resulting data set {4,4,4,6,8,8,8} has two modes (4 and 8), violating the uniqueness condition.
- Case 4: $x_3 \le 3 \Rightarrow x_1 + x_2 + x_3 \le 3 + 3 + 3 = 9$, which cannot equal 12.

Thus, there are exactly 5 distinct valid data sets, making statement (A) true.

The range of the data set is $x_7 - x_1 = 8 - x_1$. The minimum value of x_1 is 1 (from the first data set), so the maximum range is $8 - 1 = 7$. Thus, statement (B) is true.

From the valid data sets {2, 5, 5, 6, 8, 8, 8} and {3, 4, 5, 6, 8, 8, 8}, the number of observations strictly less than the mean (6) is 3. Thus, statement (C) is true.

The least possible value of x_3 among the valid sets is 5. Thus, statement (D) is false.

Paragraph

Two players, A and B, alternately roll a fair six-sided die, starting with player A. The game finishes as soon as either player rolls a '6'. Let N denote the total number of rolls made in the game. After the game concludes, a biased coin with the probability of showing heads equal to $\frac{1}{5}$ is tossed exactly N times. Let X denote the total number of heads obtained from these N coin tosses.

Q1. The probability that no heads are obtained (X = 0) is :

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{1}{2}$

Ans. B

Sol. The total number of rolls N follows a geometric distribution with success probability $p = \frac{1}{6}$ for rolling a '6'.

Thus, $P(N = n) = \left(\frac{5}{6}\right)^{n-1}\left(\frac{1}{6}\right)$ for $n \ge 1$.

Given N = n, X follows a binomial distribution $B\left(n, \frac{1}{5}\right)$.

Using the law of total probability for X = 0 :

$$P(X = 0) = \sum_{n=1}^{\infty} P(N = n)P(X = 0 \mid N = n)$$

$$P(X = 0) = \sum_{n=1}^{\infty} \left(\frac{5}{6}\right)^{n-1}\left(\frac{1}{6}\right)\binom{n}{0}\left(\frac{1}{5}\right)^{0}\left(\frac{4}{5}\right)^{n}$$

$$P(X=0)=\frac{1}{6}\cdot\frac{6}{5}\sum_{n=1}^{\infty}\left(\frac{5}{6}\cdot\frac{4}{5}\right)^n=\frac{1}{5}\sum_{n=1}^{\infty}\left(\frac{4}{6}\right)^n=\frac{1}{5}\cdot\frac{2/3}{1-2/3}=\frac{2}{5}$$

Q2. Given that exactly 2 heads are obtained (X = 2), the conditional probability that player A won the die-rolling game is :

(A) $\frac{31}{125}$ (B) $\frac{62}{125}$ (C) $\frac{63}{125}$ (D) $\frac{19}{50}$

Ans. B

Sol. For x ≥ 1, we can simplify the general term for P(X = x) :

$$P(X=x)=\sum_{n=x}^{\infty}\left(\frac{5}{6}\right)^{n-1}\left(\frac{1}{6}\right)\binom{n}{x}\left(\frac{1}{5}\right)^x\left(\frac{4}{5}\right)^{n-x}=\frac{3}{5}\left(\frac{1}{2}\right)^x$$

For X = 2, $P(X=2)=\frac{3}{5}\left(\frac{1}{2}\right)^2=\frac{3}{20}$.

Player A wins if the game ends on an odd roll (N = 2k + 1 for k ≥ 1, since N ≥ X = 2).

$$P(X=2 \text{ and } N \text{ is odd}\})=\sum_{k=1}^{\infty}P(N=2k+1)P(X=2\mid N=2k+1)$$

$$=\sum_{k=1}^{\infty}\left(\frac{5}{6}\right)^{2k}\left(\frac{1}{6}\right)\binom{2k+1}{2}\left(\frac{1}{5}\right)^2\left(\frac{4}{5}\right)^{2k-1}=\frac{1}{120}\sum_{k=1}^{\infty}\binom{2k+1}{2}\left(\frac{4}{9}\right)^k$$

Using the negative binomial summation identity $\sum_{k=1}^{\infty}\binom{2k+1}{2}t^k=\frac{t(3+t)}{(1-t)^3}$ with $t=\frac{4}{9}$:

$$\sum_{k=1}^{\infty}\binom{2k+1}{2}\left(\frac{4}{9}\right)^k=\frac{\frac{4}{9}\left(3+\frac{4}{9}\right)}{\left(1-\frac{4}{9}\right)^3}=\frac{1116}{125}$$

$$P(X=2 \text{ and } N \text{ is odd})=\frac{1}{120}\times\frac{1116}{125}=\frac{93}{1250}$$

By Bayes' theorem, the required conditional probability is:

$$P(N \text{ is odd}\mid X=2)=\frac{P(X=2 \text{and} N \text{isodd})}{P(X=2)}=\frac{93/1250}{3/20}=\frac{62}{125}$$

Q1. Let S be the set of all shortest paths from the origin $(0,0)$ to the point $(3,3)$ in the Cartesian plane, where each step is either 1 unit to the right or 1 unit upwards along the grid lines. If N denotes the number of such paths in S that pass through neither $(1,1)$ nor $(2,2)$, then find the value of N.

Ans. 4

Sol. The total number of paths from $(0,0)$ to $(3,3)$ without any restriction is:

$$T=\binom{3+3}{3}=\binom{6}{3}=20$$

Let A be the event that a path passes through $(1,1)$. The number of such paths is:

$$n(A)=\binom{1+1}{1}\times\binom{2+2}{2}=2\times 6=12$$

Let B be the event that a path passes through $(2,2)$. The number of such paths is:

$$n(B)=\binom{2+2}{2}\times\binom{1+1}{1}=6\times 2=12$$

The number of paths passing through both $(1,1)$ and $(2,2)$ is:

$$n(A \cap B) = \binom{1+1}{1} \times \binom{1+1}{1} \times \binom{1+1}{1} = 2 \times 2 \times 2 = 8$$

By the Principle of Inclusion-Exclusion, the number of paths passing through $(1,1)$ or $(2,2)$ is:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 12 + 12 - 8 = 16$$

The number of paths that pass through neither point is:

$$N = T - n(A \cup B) = 20 - 16 = 4$$

Final Answer: 4

Q2. Let $L = \lim\limits_{x \to 0} \dfrac{2\max\{x, 3x\}\min\{x^2, 2x^2\} + \max\{0, x^3\} + 5\min\{0, x^3\}}{x^2 \ln(1+x)}$. Then the value of L is equal to _____.

Ans. 7

Sol. To evaluate the limit L, we analyze the left-hand limit (LHL) and the right-hand limit (RHL) separately.

Case 1: For $x > 0$

$$\max\{x, 3x\} = 3x$$

$$\min\{x^2, 2x^2\} = x^2$$

$$\max\{0, x^3\} = x^3$$

$$\min\{0, x^3\} = 0$$

The right-hand limit is:

$$\text{RHL} = \lim_{x \to 0^+} \frac{2(3x)(x^2) + x^3 + 0}{x^2 \ln(1+x)} = \lim_{x \to 0^+} \frac{7x^3}{x^2 \cdot x} = 7$$

Case 2: For $x < 0$

$$\max\{x, 3x\} = x$$

$$\min\{x^2, 2x^2\} = x^2$$

$$\max\{0, x^3\} = 0$$

$$\min\{0, x^3\} = x^3$$

The left-hand limit is:

$$\text{LHL} = \lim_{x \to 0^-} \frac{2(x)(x^2) + 0 + 5x^3}{x^2 \ln(1+x)} = \lim_{x \to 0^-} \frac{7x^3}{x^2 \cdot x} = 7$$

Since $\text{LHL} = \text{RHL} = 7$, the value of L is 7.

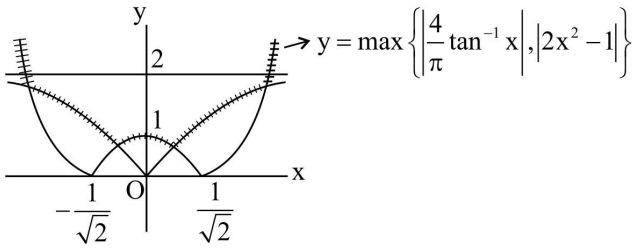
Q3. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \max\left\{\left|\frac{4}{\pi}\tan^{-1} x\right|, \left|2x^2 - 1\right|\right\}$$

If n is the total number of points in $\mathbb{R}$ where $f(x)$ is not differentiable, then the value of n is _____.

Ans. 4

Sol.



four corner points are present in the graph.

Q4. Let P be a point on the parabola $y^2 = 4x$ and Q be a point on the circle $x^2 + y^2 - 6x + 8 = 0$. If the distance PQ is minimum and C is the center of the circle, then the value of $(PC)^2$ is equal to

Ans. 8

Sol. The given equation of the circle can be rewritten as:

$$(x - 3)^2 + y^2 = 1$$

This yields the center $C(3, 0)$ and radius $r = 1$. The shortest distance between a parabola and a circle lies along the common normal, which necessarily passes through the center of the circle C. Let any parametric point on the parabola $y^2 = 4x$ be $P(t^2, 2t)$. The square of the distance from P to the center $C(3, 0)$ is:

$$(PC)^2 = (t^2 - 3)^2 + (2t - 0)^2 = t^4 - 2t^2 + 9$$

To minimize $(PC)^2$, differentiate with respect to t:

$$\frac{d}{dt}(PC)^2 = 4t^3 - 4t = 0 \implies t = 0 \text{ or } t = \pm 1$$

Evaluating the distance for these critical points: • For $t = 0$, $(PC)^2 = 9$ • For $t = \pm 1$, $(PC)^2 = 1 - 2 + 9 = 8$ Since $8 < 9$, the minimum value of $(PC)^2$ is 8.

Q5. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function satisfying the functional equation

$$f(x + y) = e^x f(y) + e^y f(x)$$

for all $x, y \in \mathbb{R}$. If the straight line $y = 4ex - 2e$ is tangent to the curve $y = f(x)$ at $x = 1$, then the value of $f^{(4)}(0) - f^{(3)}(0)$ is equal to _____ (where $f^{(n)}(x)$ denotes the n-th derivative of $f(x)$ with respect to x).

Ans. 2

Sol. $f(x + y) = e^x f(y) + e^y f(x)$

Divde both sides by e^{x+y}

$\frac{f(x+y)}{e^{x+y}} = \frac{f(x)}{e^x} + \frac{f(y)}{e^y}$ .....(1)

Now define $g(t) = \frac{f(t)}{e^t}$

use in (i)

$g(x + y) = g(x) + g(y)$.

$\Rightarrow g(t) = kt$

$\Rightarrow f(t) = kte^t$

$\because$ f'(1) = 4e (Slope of tangent at x = 1)

$\Rightarrow f'(t) = k(1 + t)e^t$

$\Rightarrow k = 2$

$f(t) = 2\ te^t$

$\Rightarrow f^{(n)}(t) = 2e^t (t + n)$

Q1. Consider the following continuous frequency distribution where the frequency k of the class interval $20 - 30$ is missing:

Class Interval	**Frequency**
$0 - 10$	6
$10 - 20$	10
$20 - 30$	k
$30 - 40$	8
$40 - 50$	4

If the median of this distribution is 23, then the value of k is _____.

Ans. 10

Sol. Since the given median is 23, it lies in the interval $20 - 30$. Thus, the median class is $20 - 30$.

From the distribution table, we identify the following parameters:

- Lower limit of the median class, $L = 20$
- Frequency of the median class, $f = k$
- Class width, $h = 10$
- Cumulative frequency of the class preceding the median class, $CF = 6 + 10 = 16$
- Total frequency, $N = 6 + 10 + k + 8 + 4 = 28 + k$

The median for a continuous frequency distribution is given by the formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f} \right) \times h$$

Substituting the respective values into the formula:

$$23 = 20 + \left(\frac{\frac{28+k}{2} - 16}{k} \right) \times 10$$

$$3 = \left(\frac{14 + \frac{k}{2} - 16}{k} \right) \times 10$$

$$3 = \left(\frac{\frac{k}{2} - 2}{k} \right) \times 10$$

$$3 = \frac{5k - 20}{k}$$

$$3k = 5k - 20$$

$$2k = 20 \implies k = 10$$

Q2. Let $S = \{1, 2, 3, \ldots, 10\}$. Find the number of subsets X of S such that X contains exactly 4 elements, no two elements in X are consecutive integers, and the product of all elements in X is a multiple of 3.

Ans. 27

Sol. The total number of ways to choose k non-consecutive elements from a set of n elements is given by the formula $\binom{n-k+1}{k}$.

Here, $n = 10$ and $k = 4$, so the total number of subsets X with no two consecutive elements is:

$$\text{Total} = \binom{10-4+1}{4} = \binom{7}{4} = 35$$

We need the product of the elements in X to be a multiple of 3. We can find this by subtracting the number of valid subsets that contain no multiples of 3 from the total.

The multiples of 3 in S are $\{3, 6, 9\}$. Excluding these, the remaining available elements are:

$$S' = \{1, 2, 4, 5, 7, 8, 10\}$$

We must choose 4 elements from S' such that no two are consecutive in the original set S. Notice that S' can be grouped into pairs of consecutive integers and one isolated integer:

$$\{1, 2\}, \{4, 5\}, \{7, 8\}, \text{ and } \{10\}$$

To choose 4 elements without picking any two that are consecutive, we can select at most one element from each of the three pairs. Thus, to get 4 elements in total, we must definitely choose the element 10, and exactly one element from each of the three pairs $\{1, 2\}$, $\{4, 5\}$, and $\{7, 8\}$.

The number of ways to make these choices is:

$$\text{Complement} = \binom{2}{1} \times \binom{2}{1} \times \binom{2}{1} \times 1 = 2 \times 2 \times 2 = 8$$

Thus, the required number of subsets is:

$$35 - 8 = 27$$

Q3. Let $A(x) = \sum_{r=0}^{10} \binom{10}{r} x^r (1+x)^r$ and $B(x) = \sum_{k=0}^{10} \binom{10}{k} 2^{10-k} (-1)^k (1+x)^k$ be two polynomial expressions. If the coefficient of x^6 in the product $A(x) \cdot B(x)$ is C, then the value of C is _____.

Ans. 45

Sol. Using the Binomial Theorem, the first expression $A(x)$ can be simplified by substituting $y = x(1+x) = x + x^2$:

$$A(x) = \sum_{r=0}^{10} \binom{10}{r} (x + x^2)^r = (1 + x + x^2)^{10}$$

Similarly, the second expression $B(x)$ can be simplified by rewriting the summation terms:

$$B(x) = \sum_{k=0}^{10} \binom{10}{k} 2^{10-k} (-(1+x))^k = (2 - (1+x))^{10} = (1-x)^{10}$$

Now, consider the product $A(x) \cdot B(x)$:

$$A(x) \cdot B(x) = (1 + x + x^2)^{10} \cdot (1-x)^{10} = \left[(1-x)(1+x+x^2)\right]^{10} = (1 - x^3)^{10}$$

The general term in the expansion of $(1 - x^3)^{10}$ is given by:

$$T_{m+1} = \binom{10}{m} (-1)^m (x^3)^m = \binom{10}{m} (-1)^m x^{3m}$$

To find the coefficient of x^6, we set the exponent of x equal to 6:

$$3m = 6 \implies m = 2$$

Substituting $m = 2$ back into the general term yields the coefficient C:

$$C = \binom{10}{2} (-1)^2 = \frac{10 \times 9}{2} \times 1 = 45$$